

Scalability & Future Plan

Date	10 july 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only text-based emotion detection	Cannot analyze speech or facial expressions	High
2	Single-user deployment	Limited scalability for multiple concurrent users	Medium
3	Limited emotion categories and recommendation dataset	Reduces personalization for diverse learning need	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	User authentication, dashboard, and prediction history	Better user experience and secure access
2	Data Storage	Voice emotion recognition using speech processing	Supports multimodal emotion detection
3	Performance	Facial emotion recognition using webcam integration	Higher prediction accuracy through visual analysis
4	Security	Basic API key protection	Implement JWT authentication, HTTPS, encrypted storage, and role-based access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	User authentication, dashboard, and prediction history	1–2 Month	Better user experience and secure access
Phase 3	Voice emotion recognition using speech processing	3–4 Months	Higher prediction accuracy through visual analysis
Phase 4	Personalized AI tutor, multilingual support, and mobile application	6–12 Months	Wider accessibility, intelligent learning assistance, and improved scalability